Principal Component Analysis All in One

Version 0.7.0

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Contents

1 Introduction **3**

2 Features **3**

3 What's new in v0.7.0 **4**

4 Cite the work **4**

5 Quick start **4**

6 Installation **6**
6.1 Download compiled binary . . . 6
6.2 Via Conda . . . 6
6.3 Build from source . . . 6

7 Documentation **8**
7.1 Options . . . 8
7.2 Which SVD method to use . . . 10
7.3 Input formats . . . 10
7.4 Output files . . . 10
7.5 Performance and memory . . . 11
7.6 Data Normalization . . . 11
7.7 Projection . . . 11
7.8 Genome-wide selection scan . . . 12
7.9 HWE accounting for population structure . . . 12
7.10 Ancestry adjusted LD matrix . . . 13
7.11 Report LD statistics . . . 13
7.12 Prunning based on ancestry adjusted LD . . . 13
7.13 Clumping based on ancestry adjusted LD . . . 13

8 More tutorials **13**
8.1 Genotype data (PLINK) . . . 14
8.2 PLINK2 PGEN input . . . 14
8.3 BGEN input (limited support) . . . 14
8.4 Single cell RNA-seq data (CSV) . . . 14

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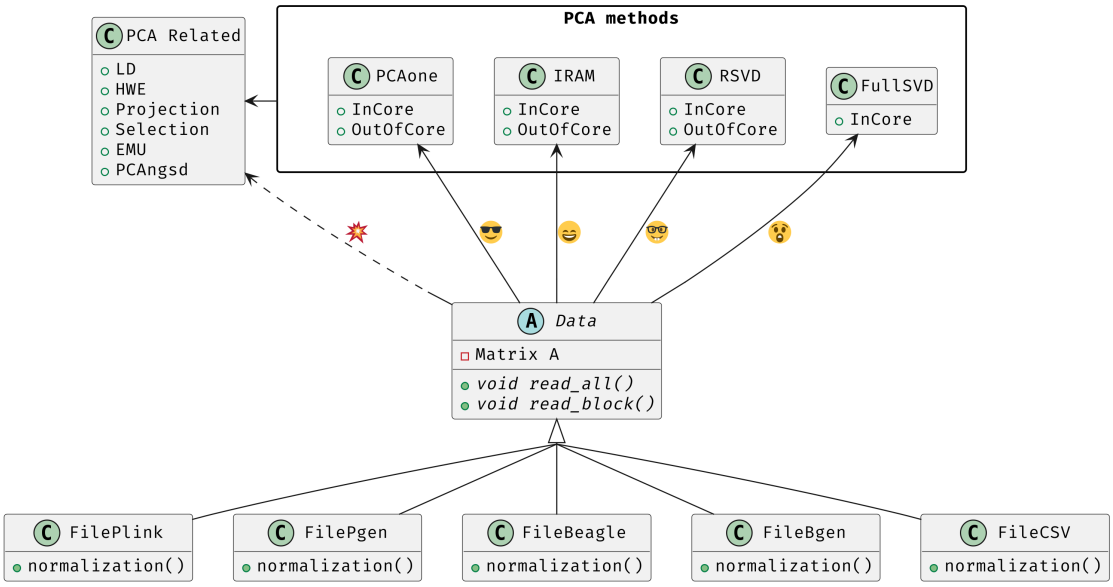
9 Acknowledgements **14**

10 Contributing **15**

1 Introduction

PCAone is a fast and memory efficient PCA tool implemented in C++ aiming at providing comprehensive features and algorithms for different scenarios. PCAone implements 3 fast PCA algorithms for finding the top eigenvectors of large datasets, which are Implicitly Restarted Arnoldi Method (IRAM, –svd 0), single pass Randomized SVD with power iteration scheme (sSVD, –svd 1, Algorithm1 in paper) and **our proposed window based RSVD** (winSVD, –svd 2, Algorithm2 in paper). All have both in-core and out-of-core implementation. Addtionally, Full SVD/Eigendecomposition (–svd 3) is supported via in-core mode only. Also, check out the R package here. PCAone supports multiple different input formats, such as PLINK BED, PLINK2 PGEN, BGEN, Beagle genetic data formats and a general comma separated CSV format for other data, such as scRNAs and bulk RNAs. For genetics data, PCAone also implements EMU and PCAngsd algorithm for data with missingness and uncertainty. The PDF manual can be downloaded here.



2 Features

See change log here.

- Has both Implicitly Restarted Arnoldi Method (IRAM) and Randomized SVD (RSVD) with **out-of-core** implementation.
- Implements our new fast window based Randomized SVD algorithm for tera-scale dataset.
- Quite fast with multi-threading support using high performance library MKL or OpenBLAS as backend.
- Supports the PLINK, BGEN, Beagle genetic data formats.
- Supports PLINK2 PGEN input via `--pgen`, using dosages by default and hard calls with `--hardcall`.
- Supports the comma separated CSV format with integer/float entries for single cell (or bulk) RNA-seq data compressed by zstd.
- Supports EMU algorithm for scenario with large proportion of missingness.

- Supports PCAngsd algorithm for low coverage sequencing scenario with genotype likelihood as input.
- Novel LD prunning and clumping method that accounts for population structure in the data.
- Projection support for data with missingness.
- Projection from BEAGLE genotype likelihoods onto a reference space via `--project 3`.
- HWE test taking population structure into account.

3 What's new in v0.7.0

- Added PLINK2 `PGEN` input support with `--pgen` for PCA workflows.
- `PGEN` reads dosages when available, and can be forced to use hard calls with `--hardcall`.
- Added genotype-likelihood aware projection for `BEAGLE` input with `--project 3`.
- Projection now matches overlapping markers against the reference `.mbim` and corrects flipped alleles when needed.
- Added genome-wide selection scans via `--selection 1` (Galinsky/FastPCA) and `--selection 2` (pcadapt).
- Bug fixes for projection with missing data
- Fast Eigendecomposition when `N<<M` with `-d 3`

4 Cite the work

- If you use PCAone, please first cite our paper on genome reseach Fast and accurate out-of-core PCA framework for large scale biobank data.
- If using the EMU algorithm, please also cite Large-scale inference of population structure in presence of missingness using PCA.
- If using the PCAngsd algorithm, please also cite Inferring Population Structure and Admixture Proportions in Low-Depth NGS Data.
- If using the ancestry ajusted LD statistics for pruning and clumping, please also cite Measuring linkage disequilibrium and improvement of pruning and clumping in structured populations.

5 Quick start

```
pkg=https://github.com/Zilong-Li/PCAone/releases/latest/download/PCAone-avx2-Linux.zip
wget $pkg && unzip -o PCAone-avx2-Linux.zip
wget http://popgen.dk/zilong/datahub/pca/example.tar.gz
tar -xzf example.tar.gz && rm -f example.tar.gz
./PCAone -b example/plink
R -s -e 'd=read.table("pcaone.eigvecs2", h=F);
plot(d[,1:2+2], col=factor(d[,1]), xlab="PC1", ylab="PC2");
legend("topleft", legend=levels(factor(d[,1])), col=1:4, pch = 21, cex=1.2);'
```

You will find these files in current directory.

```
PCAone            # program
Rplots.pdf        # pca plot
example           # folder of example data
pcaone.eigvals    # eigenvalues
pcaone.eigvecs    # eigenvectors, the PCs you need to plot
pcaone.eigvecs2   # eigenvectors with header line
pcaone.log        # log file
```

6 Installation

There are 3 ways to install PCAone.

6.1 Download compiled binary

There are compiled binaries provided for both Linux and Mac platform. Check the releases page to download one.

```
pkg=https://github.com/Zilong-Li/PCAone/releases/latest/download/PCAone-Linux.zip
wget $pkg || curl -LO $pkg
unzip -o PCAone-Linux.zip
```

6.2 Via Conda

PCAone is also available from bioconda.

```
conda config --add channels bioconda
conda install pcaone
PCAone --help
```

6.3 Build from source

PCAone has been tested on both `Linux` and `MacOS` system. To build PCAone from the source code, the following dependencies are required:

- GCC/Clang compiler with C++17 support
- GNU make
- zlib

On Linux, we **recommend** building the software from source with MKL as backend to maximize the performance.

6.3.1 With MKL or OpenBLAS as backend

Build PCAone dynamically with MKL can maximize the performance for large dataset particularly, because the faster threading layer `libiomp5` will be linked at runtime. There are two options to obtain MKL library:

- download `MKL` from the website

After having `MKL` installed, find the `MKL` root path and replace the path below with your own.

```
make -j4 MKLROOT=/opt/intel/oneapi/mkl/latest  ONEAPI_COMPILER=/opt/intel/oneapi/compiler/latest
```

Alternatively, for advanced user, modify variables directly in `Makefile` and run `make` to use MKL or OpenBlas as backend.

- install `MKL` by conda

```
conda install -c conda-forge -c anaconda -y mkl mkl-include intel-openmp
git clone https://github.com/Zilong-Li/PCAone.git
cd PCAone
# if mkl is installed by conda then use ${CONDA_PREFIX} as mklroot
make -j4 MKLROOT=${CONDA_PREFIX}
./PCAone -h
```

6.3.2 Without MKL or OpenBLAS dependency

If you don't want any optimized math library as backend, just run:

```
git clone https://github.com/Zilong-Li/PCAone.git
cd PCAone
make -j4
./PCAone -h
```

6.3.3 For MacOS users, check out the mac workflow.

```
brew install libomp
make -j4
```

7 Documentation

7.1 Options

Run `PCAone --groff > pcaone.1 && man ./pcaone.1` or `PCAone --help` to read the manual. Here are common options.

```
General options:
  -h, --help                    print all options including hidden advanced options
  -m, --memory arg (=0)         RAM usage in GB unit for out-of-core mode. default is in-core
  ↪  mode
  -n, --threads arg (=12)       the number of threads to be used
  -v, --verbose arg (=1)        verbosity level for logs. Options are
                                0: silent, no messages on screen;
                                1: concise messages to screen;
                                2: more verbose information;
                                3: enable debug information.

PCA algorithms:
  -d, --svd arg (=2)            SVD method to be applied. default 2 is recommended for big data.
  ↪  Options are
                                0: the Implicitly Restarted Arnoldi Method (IRAM);
                                1: the Yu's single-pass Randomized SVD with power iterations;
                                2: the accurate window-based Randomized SVD method (PCAone);
                                3: the full Singular Value Decomposition.
  -k, --pc arg (=10)            top k principal components (PCs) to be calculated
  -C, --scale arg (=-9)         do normalization or scaling for input file. Options are
                                -9: standardize genetic data by sqrt(ploidy*f*(1-f));
                                 0: do nothing and proceed to SVD;
                                 1: do direct standardization, as the scale(x, center=TRUE,
                                 ↪  scale=TRUE) function in R;
                                 2: do first count per median log transformation (CPMED), then
                                 ↪  standardization;
                                 3: do first log1p transformation, then standardization;
                                 4: do first relative counts, then standardization.
  --maxp arg (=20)              maximum number of power iterations for RSVD algorithm.
  -S, --no-shuffle              do not shuffle columns of data for --svd 2 (if not locally
  ↪  correlated).
  --seed arg (=112)             seeds for reproducing results.
  --emu                         use EMU algorithm for genotype input with missingness.
  --pcangsd                     use PCAngsd algorithm for genotype likelihood input.

Input options:
  -b, --bfile arg               prefix of PLINK .bed/.bim/.fam files.
  -p, --pgen arg                prefix of PLINK2 .pgen/.pvar/.psam files.
  -B, --binary arg              path of binary file.
  -c, --csv arg                 path of comma seperated CSV file compressed by zstd.
  -g, --bgen arg                path of BGEN file compressed by gzip/zstd.
  -G, --beagle arg              path of BEAGLE file compressed by gzip.
  -F, --match-bim arg           the .mbim file to be matched, where the 7th column is allele
  ↪  frequency.
  -P, --USV arg                 prefix of PCAone .eigvecs/.eigvals/.loadings/.mbim.

Output options:
  -o, --out arg (=pcaone)       prefix of output files. default [pcaone].
  -V, --printv                  output the right eigenvectors with suffix .loadings.
  -D, --ld                      output a binary matrix for downstream LD related analysis.
  -R, --print-r2                print LD R2 to *.ld.gz file for pairwise SNPs within a window
  ↪  controlled by --ld-bp.

Misc options:
  --maf arg (=0)                exclude variants with MAF lower than this value
  --project arg (=0)            project the new samples onto the existing PCs. Options are
```

```
                                0: disabled;
                                1: by multiplying the loadings with mean imputation for missing
                                ↪  genotypes;
                                2: by solving the least squares system Vx=g. skip sites with
                                ↪  missingness;
                                3: by EM to account for genotype uncertainty (BEAGLE input).
--inbreed arg (=0)              compute the inbreeding coefficient accounting for population
↪  structure. Options are
                                0: disabled;
                                1: compute per-site inbreeding coefficient and HWE test.
--selection arg (=0)            compute selection statistics. Options are
                                0: disabled;
                                1: perform genome-wide selection scan using Galinsky et al
                                ↪  method;
                                2: perform genome-wide selection scan using pcadapt method.
--ld-r2 arg (=0)                R2 cutoff for LD-based pruning (usually 0.2).
--ld-bp arg (=1000000)          physical distance threshold in bases for LD window.
--ld-stats arg (=0)             statistics to compute LD R2 for pairwise SNPs. Options are
                                0: the ancestry adjusted, i.e. correlation between residuals;
                                1: the standard, i.e. correlation between two alleles.
--clump arg                     assoc-like file with target variants and pvalues for clumping.
--clump-names arg (=CHR,BP,P)   column names in assoc-like file for locating chr, pos and
↪  pvalue.
--clump-p1 arg (=0.0001)        significance threshold for index SNPs.
--clump-p2 arg (=0.01)          secondary significance threshold for clumped SNPs.
--clump-r2 arg (=0.5)           r2 cutoff for LD-based clumping.
--clump-bp arg (=250000)        physical distance threshold in bases for clumping.
```

7.2 Which SVD method to use

This depends on your datasets, particularlly the relationship between number of samples (`N`) and the number of variants / features (`M`) and the top PCs (`k`). Here is an overview and the recommendation.

Method	Scenario	Accuracy	Speed
Full SVD (-d 3)	full variance explained	Exact	slow for big `N` and `M`
winSVD (-d 2)	`M or N >> 500000`	Very high	fast (only 7 iterations used)
IRAM (-d 0)	speed insensitive	Very high	denpends on `N` and # iterations
sSVD (-d 1)	accuracy insensitive	High	depends on # iterations

7.3 Input formats

PCAone is designed to be extensible to accept many different formats. Currently, PCAone can work with SNP major genetic formats to study population structure. such as PLINK, BGEN and Beagle. Also, PCAone supports a comma delimited CSV format compressed by zstd, which is useful for other datasets requiring specific normalization such as single cell RNAs data. Since `v0.7.0`, PCAone also supports PLINK2 PGEN input via `--pgen`. If dosages are stored in the `.pgen` file, PCAone uses them by default; add `--hardcall` to force hard-call genotypes instead. The current `BGEN` support is limited, so for large production workflows we recommend converting `BGEN` to `PGEN` when possible.

7.4 Output files

7.4.1 Eigen vectors

Eigen vectors are saved in file with suffix `.eigvecs`. Each row represents a sample and each col represents a PC.

7.4.2 Eigen/Singular values

Eigenvalues and signularvalues are saved in file with suffix `.eigvals` and `.sigvals` respectively. Each row represents the eigenvalue/singularvalue of corresponding PC.

7.4.3 Features loadings

Features Loadings are saved in file with suffix `.loadings`. Each row represents a feature and each column represents a corresponding PC. Use `--printv` option to output it.

7.4.4 Variants infomation

A plink-like bim file named with `.mbim` is used to store the variants list with extra information. Currently, the `mbim` file has 7 columns with the 7th being the allele frequency. PCAone writes this file automatically whenever it outputs `.loadings` via `--printv`, and for LD-related outputs that need variant metadata downstream.

7.4.5 LD matrix

The matrix for calculating the ancestry-adjusted LD is saved in a file with suffix `.residuals`, and its associated variants information is stored in `mbim` file. For the binary file, the first 4-bytes stores the number of variants/SNPs, and the second 4-bytes stores the number of samples in the matrix. Then, the rest of the file is a sequence of **M** blocks of **N x 4** bytes each, where **M** is the number of variants and **N** is the number of samples. The first block corresponds to the first marker in the `.mbim` file, etc.

7.4.6 LD R2

The LD R2 for pairwise SNPs within a window can be outputted to a file with suffix `ld.gz` via `--print-r2` option. This file uses the same long format as the one plink used.

7.5 Performance and memory

PCAone has both **in-core** and **out-of-core** mode for 3 different partial SVD algorithms, which are IRAM (`--svd 0`), sSVD (`--svd 1`) and winSVD (`--svd 2`). Also, PCAone supports Full SVD (`--svd 3`) but with only **in-core** mode. Therefore, there are **7** ways for doing PCA in PCAone. In default PCAone uses **in-core** mode, which is the fastest way (**NOTE**: you can gain some speedup for in-code computation by limiting the `-n threads` to half of the available threads of your machine). However, in case the server runs out of memory, you can trigger `out-of-core mode` by specifying the amount of memory using `-m/--memory` option with a value greater than 0. Normally, use `-m 1` is enough for large dataset and PCAone will allocate more RAM when needed.

7.5.1 Run winSVD method (default) with in-core mode

```
./PCAone --bfile example/plink
```

7.5.2 Run winSVD method with out-of-core mode

```
./PCAone --bfile example/plink -m 2
```

7.5.3 Run sSVD method with out-of-core mode

```
./PCAone --bfile example/plink --svd 1 -m 2
```

7.5.4 Run IRAM method with out-of-core mode

```
./PCAone --bfile example/plink --svd 0 -m 2
```

7.5.5 Run Full SVD method with in-core mode

```
./PCAone --bfile example/plink --svd 3
```

7.6 Data Normalization

Note that data will be always first center for PCA. Additionally, there are several different normalization method implemented with `--scale` option. One should choose proper normalization method for specific type of data. In default, PCAone will automatically apply the standard normalization for genetic data.

7.7 Projection

Project new samples onto existing PCs is supported with `--project` option. First, we run PCAone on a set of reference samples and output the loadings:

```
PCAone -b example/ref -k 10 --printv -o ref
```

Then, we need to read in the SNPs loadings from the ref set (`--read-V`) and its scaling factors (`--read-S`) as well as the allele frequencies form the `.mbim` file via `--match-bim`. You can use the `--USV` option instead to simplify the usage since **v0.4.8**. For projection, PCAone matches overlapping markers against the reference `.mbim` file and can correct flipped alleles when needed, but you should still make sure both datasets are aligned to the same genome build and alleles.

Below is a command using example PLINK data to project new target samples onto the reference coordinates.

```
PCAone -b example/new \
       --USV ref \      ## prefix to .eigvecs, .eigvals, .loadings, .mbim
       --project 2 \    ## check the manual on projection methods
       -o new
```

For BEAGLE genotype likelihood input, use `--project 3` to perform genotype-likelihood aware projection with an EM procedure:

```
PCAone -G example/target.beagle.gz \
       --USV ref \
       --project 3 \
       --scale 0 \
       -o target
```

7.8 Genome-wide selection scan

PCAone can test for differentiated variants after a good reference PCA has been computed with good sites after site-level QC, e.g. LD prunning.

First run PCA on the subset genotype matrix with good sites

```
PCAone -b example/subset -k 10 -o goodsites
```

Then reuse the saved `USV` outputs with `--selection`. Method `1` computes the Galinsky/FastPCA statistic for each variant and PC, while method `2` computes pcadapt-style z-scores, chi-square statistics, p-values, and a genomic inflation factor.

```
PCAone -b example/fullset \
       --USV goodsites \
       --selection 1 \
       -o sel
```

```
PCAone -b example/fullset \
       --USV goodsites \
       --selection 2 \
       -o sel
```

The selection outputs are written per variant in the same marker order as the plink `.bim` file.

7.9 HWE accounting for population structure

To test Hardy-Weinberg equilibrium in presence of population structure like pcangsd, we need to work on the so-called individual allele frequencies matrix Π, which can be reconstructed via the output of PCAone, i.e the `.eigvecs,.sigvals,.loadings and .mbim` files, generated by first running

```
PCAone -b example/plink -k 3 -V -o pcaone --pcangsd
## alternative for PLINK input with missingness
## PCAone -b example/plink -k 3 -V -o pcaone --emu
```

Then we apply `--inbreed 1` option to obtain the P value of HWE and inbreeding coefficient per-site. The per-site statistics is outputted to a file with suffix `.hwe`.

```
PCAone -b example/plink \
       --USV pcaone \
       --inbreed 1 \
       -o inbreed
```

7.10 Ancestry adjusted LD matrix

LD patterns vary across diverse ancestry and structured groups, and conventional LD statistics, e.g. the implementation in `plink --ld`, failed to model the LD in admixed populations. Thus, we can use the so-called ancestry-adjusted LD statistics to account for population structure in LD. See our paper for more details.

To calculate the ancestry-adjusted LD matrix, we first figure out the number of principal components (`-k/--pc`) that capture population structure. In this example, assuming that 3 PCs can accout for population structure, we enable `--ld` option to calculate and output the ancestry adjusted LD matrix in a file with suffix `.residuals`.

```
./PCAone -b example/plink -k 3 --ld -o adj
```

7.11 Report LD statistics

Currently, the LD R2 for pairwise SNPs within a window can be outputted via `--print-r2` option.

```
./PCAone -B adj.residuals \
         --match-bim adj.mbim \
         --ld-bp 1000000 \
         --print-r2 \
         -o adj
```

We provide the calc_decay_bin.R script to parse the output file `.ld.gz` and calculate the average R2 for each bin as well as plotting. We also provide the nextflow ld.nf for benchmarking the LD statistics.

7.12 Prunning based on ancestry adjusted LD

Given the LD binary file `.residuals` and its associated variant file `.mbim`, we can do pruning based on user-defined thresholds and windows

```
./PCAone -B adj.residuals \
         --match-bim adj.mbim \
         --ld-r2 0.8 \
         --ld-bp 1000000 \
         -o adj
```

7.13 Clumping based on ancestry adjusted LD

Likewise, we can do clumping based on the Ancestry-Adjusted LD matrix and user-defined association results

```
./PCAone -B adj_ld.residuals \
         --match-bim adj.mbim \
         --clump example/plink.pheno0.assoc,example/plink.pheno1.assoc  \
         --clump-p1 0.05 \
         --clump-p2 0.01 \
         --clump-r2 0.1 \
         --clump-bp 10000000 \
         -o adj
```

8 More tutorials

Let's download the example data first if you haven't done so.

```
wget http://popgen.dk/zilong/datahub/pca/example.tar.gz
tar -xzf example.tar.gz && rm -f example.tar.gz
```

8.1 Genotype data (PLINK)

If you want to get the variance explained of each PC, we need to use `--svd 3` to run Full SVD.

```
./PCAone --bfile example/plink -d 3
```

Then, we can make a PCA plot in R.

```
pcs <- read.table("pcaone.eigvecs2",h=F)
vals <- scan("pcaone.eigvals")
vals <- round(vals / sum(vals), 4)
xlab <- paste0("PC1: ", vals[1] * 100, "%")
ylab <- paste0("PC2: ", vals[2] * 100, "%")
plot(pcs[,1:2+2], col=factor(pcs[,1]), xlab = xlab, ylab = ylab, cex.lab = 1.5)
legend("topleft", legend=levels(factor(pcs[,1])), col=1:4, pch = 21, cex=1.2)
```

8.2 PLINK2 PGEN input

PCAone (version >= 0.7.0) can also read PLINK2 `.pgen/.pvar/.psam` input directly. By default it uses dosages when they are present in the `.pgen` file.

```
./PCAone --pgen example/plink2_data -k 10 -m 2
```

If you want to ignore dosages and use hard-call genotypes instead, add `--hardcall`.

```
./PCAone --pgen example/plink2_data -k 10 -m 2 --hardcall
```

8.3 BGEN input (limited support)

NB: `BGEN support is very limited. Please convert BGEN to PGEN and use PGEN input!`

Imputation tools usually generate the genotype probabilities or dosages in BGEN format. To do PCA with the imputed genotype probabilities, we can work on BGEN file with `--bgen` option instead.

```
./PCAone --bgen example/test.bgen -k 10 -m 2
```

8.4 Single cell RNA-seq data (CSV)

In this example, we run PCA for the scRNAs-seq data using CSV format with a normalization method called count per median log transformation (CPMED). Since the features (genes) tend to be not correlated locally, we use `-S` option to disable permutation for winSVD.

```
./PCAone --csv example/BrainSpinalCord.csv.zst -k 10 -m 2 --scale 2 -S
```

9 Acknowledgements

PCAone uses Eigen for linear algebra operation. The IRAM method is based on yixuan/spectra. The bgen lib is ported from jeremymcrae/bgen. The EMU and PCAngsd algorithms are modified from @Jonas packages.

10 Contributing

PCAone is an open source project, and we welcome quality contributions. If you are interested in contributing and have experience with C++ (or LLM :D), feel free to reach out with ideas you would like to implement yourself.